SUPPLEMENT TO "IDENTIFYING RATIONAL TYPES IN UNKNOWN ENVIRONMENTS" APPENDIX 1: MACRO DYNAMICS — COINTEGRATION AND THE VEC FOUNDATION

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This supplement presents in full detail the macroeconometric results summarized in Section 2.2 of the main paper ("Macro dynamics: cointegration and the VEC foundation"). It documents the data, the integration-order tests, the Johansen cointegration analysis, the estimated Vector Error Correction (VEC) model with its adjustment dynamics and residual diagnostics, the Gonzalo–Granger permanent–transitory decomposition, the univariate Local Linear Trend (LLT) state-space model, and the empirical verification of the stochastic-trend equivalence corollary. All estimates were re-computed from the replication scripts cited in Section S9; rounded values use two decimal places aligned with the replication package outputs.

S1. DATA AND SAMPLE

The vector of interest is the monthly triple

$$Y_t = \left(K_t^E,\, K_t^{NE},\, r_t\right)', \tag{S1.1}$$

where K_t^E is the number of kidnappings for ransom (extortive kidnapping), K_t^{NE} is the number of non-extortive kidnappings, and r_t is the commercial lending interest rate. Kidnapping counts come from the National Center of Historical Memory (CNMH, 2013, 2026) and official statistics of the Colombian State (Ministerio de Defensa Nacional, 2026); the interest rate is the active commercial rate used as the indicator of macro-financial conditions (Masih, 1995, Suarez Padilla, 2025).

The estimation sample is the continuous monthly series from July 1954 to December 2025 ($T = 858$ observations). The figures in the main text display the period 1964–2025, which concentrates the non-trivial variation of the kidnapping series; estimation uses the full sample. Isolated missing values were imputed with a centered moving median of order three before estimation.

Economically, r_t proxies the macroeconomic environment that fixes the opportunity cost and the continuation value of crime, while K_t^{NE} marks the phase in which the criminal organization accumulates logistic capacity, operational experience, and territorial control, lowering its marginal cost of operation and enabling the more systematic organization of extortive kidnapping.

S2. INTEGRATION ORDER

Table I reports the Augmented Dickey–Fuller test of Dickey and Fuller (1979) and the KPSS test of Kwiatkowski, Phillips, Schmidt, and Shin (1992), in levels and in first differences, with

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TABLE I

AUGMENTED DICKEY–FULLER AND KPSS UNIT-ROOT TEST p-VALUES

	Levels		First differences	
Series	ADF p	KPSS p	ADF p	KPSS p
K_t^E	0.28	< 0.01	< 0.01	> 0.10
K_t^{NE}	0.41	< 0.01	< 0.01	> 0.10
r_t	0.44	< 0.01	< 0.01	> 0.10

Note. ADF: constant and 12 augmentation lags (`model = ARD`). KPSS: level-stationarity test, 12 lags, no trend. MATLAB truncates p-values at $[0.01, 0.10]$ (KPSS) and $[0.001, 0.999]$ (ADF). In levels, ADF does not reject the unit-root null at 5%; KPSS rejects stationarity at 1%. In first differences, ADF rejects the unit-root null at 1%; KPSS does not reject stationarity at 10%. *Source:* Authors' own calculations based on the monthly sample in Section S1 ($T = 858$). Software: MATLAB R2025b (The MathWorks, 2025); Econometrics Toolbox.

TABLE II

JOHANSEN TRACE TEST FOR COINTEGRATION RANK

Null hypothesis	Trace statistic	95% critical value	p-value	Decision
$r = 0$	40.32	29.80	< 0.01	Reject at 1%
$r \leq 1$	13.17	15.49	0.11	Not reject at 5%
$r \leq 2$	3.02	3.84	0.08	Not reject at 5%

Note. VAR in levels with $p = 36$ lags and unrestricted constant (MATLAB specification `H1`). Decisions follow the asymptotic p-value at the stated significance level. The null $r \leq 2$ would be rejected at 10% ($p = 0.08$); rank $r = 1$ is selected using the 5% criterion. *Source:* Authors' own calculations based on the monthly sample in Section S1 ($T = 858$). Software: MATLAB R2025b (The MathWorks, 2025); Econometrics Toolbox.

twelve augmentation lags and a constant. At the 5% level, the ADF test does not reject the unit-root null in levels for any series ($p > 0.05$), while the KPSS test rejects stationarity in levels at the 1% level ($p \leq 0.01$). In first differences, the ADF test rejects the unit-root null at the 1% level ($p = 0.001$) and the KPSS test does not reject stationarity at the 10% level ($p \geq 0.10$). These four criteria jointly classify K_t^E, K_t^{NE}, and r_t as $I(1)$ at the 5% level.

S3. JOHANSEN COINTEGRATION ANALYSIS

The cointegration rank is determined with the Johansen trace test (Johansen, 1988) on the levels VAR with $p = 36$ lags and an unrestricted constant (MATLAB deterministic specification `H1`). The lag order was selected so that the implied VEC(35) exhibits no residual autocorrelation (Section S4.3) while remaining parsimonious under the AIC among admissible specifications. Table II reports the results: the null $r = 0$ is rejected at the 1% level ($p < 0.01$), whereas $r \leq 1$ is not rejected at the 5% level ($p = 0.11$) and $r \leq 2$ is not rejected at the 5% level ($p = 0.08$) although it would be rejected at the 10% level. The evidence therefore supports exactly one cointegrating vector, $r = 1$, with $k - r = 2$ common stochastic trends.

S4. THE VEC MODEL

S4.0.0.0. *Long-run equilibrium*

With $r = 1$ and $q = p - 1 = 35$ short-run lag matrices, the estimated VEC is

$$\Delta Y_t = \alpha\, \beta' Y_{t-1} + \sum_{j=1}^{35} \Gamma_j\, \Delta Y_{t-j} + c + \varepsilon_t, \tag{S4.1}$$

estimated by maximum likelihood (Johansen reduced-rank regression). The fit statistics are $\log L = -7{,}667.23$, AIC $= 15{,}980.45$, and BIC $= 17{,}516.19$. The unnormalized cointegrating vector is

$$0.04\, K_t^{E} \;-\; 0.07\, K_t^{NE} \;-\; 0.05\, r_t \;=\; \tilde{\varepsilon}_t, \qquad \tilde{\varepsilon}_t \sim I(0), \tag{S4.2}$$

and an ADF test on the error-correction term $\tilde{\varepsilon}_t = \beta' Y_t$ rejects the unit-root null at the 1% level ($p < 0.01$), confirming the stationarity of the long-run relation. Normalizing (S4.2) on K_t^E yields the structural long-run equation

$$K_t^{E} \;=\; 1.93\, K_t^{NE} \;+\; 1.31\, r_t \;+\; \varepsilon_t, \qquad \varepsilon_t \sim I(0). \tag{S4.3}$$

The coefficient on K_t^{NE} indicates that non-extortive kidnapping operates as a signal of accumulated coercive and organizational capacity: in the long run, a unit increase in this modality is associated with an increase of approximately 1.93 units in extortive kidnapping, as the expansion of logistic infrastructure, territorial control, and intimidation lowers the marginal cost of moving toward higher-rent activities. The coefficient on the interest rate reflects a liquidity-constraint mechanism: periods of monetary contraction raise the financing cost of criminal organizations and push them toward extortive kidnapping as a source of immediate rent.

S4.0.0.0. *Adjustment dynamics*

Table III reports the estimated adjustment (loading) vector α with standard errors. Written equation by equation, the error-correction system is

$$\Delta K_t^{E} = -2.94\, \tilde{\varepsilon}_{t-1} + \textstyle\sum_{j=1}^{35} \Gamma_{1,j}\, \Delta Y_{t-j} + \varepsilon_{1,t}, \tag{S4.4}$$

$$\Delta K_t^{NE} = \quad 0.36\, \tilde{\varepsilon}_{t-1} + \textstyle\sum_{j=1}^{35} \Gamma_{2,j}\, \Delta Y_{t-j} + \varepsilon_{2,t}, \tag{S4.5}$$

$$\Delta r_t = -0.06\, \tilde{\varepsilon}_{t-1} + \textstyle\sum_{j=1}^{35} \Gamma_{3,j}\, \Delta Y_{t-j} + \varepsilon_{3,t}. \tag{S4.6}$$

Only the loading in the K^E equation is statistically significant at the 1% level ($p < 0.01$). The loading in the K^{NE} equation is not significant at the 5% or 10% level ($p = 0.17$). The loading in the r equation is not significant at the 5% level ($p = 0.10$) but is marginally significant at the 10% level (exact $p = 0.098$). Extortive kidnapping is therefore the variable that absorbs disequilibria at conventional levels, while K_t^{NE} and r_t are weakly exogenous with respect to the long-run relation at the 5% level. Because $\beta_{K^E} = 0.04$, the effective adjustment speed is

$$\left|\alpha_{K^E} \cdot \beta_{K^E}\right| \;=\; |-2.94 \times 0.04| \;\approx\; 0.11: \tag{S4.7}$$

approximately 11% of any deviation from the long-run equilibrium is corrected each month through adjustments in extortive kidnapping.

S4.0.0.0. *Residual diagnostics*

Table IV summarizes the residual diagnostics of the estimated system. Ljung–Box tests at twenty lags do not reject the absence of autocorrelation at any conventional level ($p \geq 0.99$), validating the dynamic specification with $p = 36$. Engle ARCH tests at five lags reject homoskedasticity at the 1% level in the K^E and r equations ($p < 0.01$) and do not reject it at the 5% level in the K^{NE} equation ($p = 0.95$). This pattern is expected in long monthly series spanning regimes of very different volatility (armed-conflict escalation and de-escalation, interest-rate liberalization) and does not bias the point estimates of β and α, although inference on short-run coefficients should be read with heteroskedasticity-robust caution.

TABLE III

VECTOR ERROR CORRECTION LOADING COEFFICIENTS

Equation	$\hat{\alpha}$	Std. error	p-value	Significance
ΔK_t^E	-2.94	0.67	< 0.01	1%
ΔK_t^{NE}	0.36	0.26	0.17	—
Δr_t	-0.06	0.04	0.10	10%

Note. VEC($p = 36$, $r = 1$), unrestricted constant, maximum-likelihood estimation ($T = 858$). Two-sided p-values. The Significance column reports the highest conventional level at which the coefficient differs from zero (1%, 5%, 10%); "—" denotes not significant at 10%. The exact p-value for Δr_t is 0.098. *Source:* Authors' own calculations based on the monthly sample in Section S1 ($T = 858$). Software: MATLAB R2025b (The MathWorks, 2025); Econometrics Toolbox.

TABLE IV

RESIDUAL DIAGNOSTIC TESTS FOR THE ESTIMATED VEC SYSTEM

Equation	Ljung–Box(20) p	ARCH(5) p	ARCH decision
ΔK_t^E	1.00	< 0.01	Reject at 1%
ΔK_t^{NE}	0.99	0.95	Not reject at 5%
Δr_t	1.00	< 0.01	Reject at 1%

Note. VEC($p = 36$, $r = 1$), 35 short-run lags. Ljung–Box test on VEC residuals at 20 lags; Engle ARCH LM test at 5 lags. ARCH decision: reject homoskedasticity at 1% or do not reject at 5%, as indicated. *Source:* Authors' own calculations based on the monthly sample in Section S1 ($T = 858$). Software: MATLAB R2025b (The MathWorks, 2025); Econometrics Toolbox.

S5. GONZALO–GRANGER PERMANENT–TRANSITORY DECOMPOSITION

With $k = 3$ variables and $r = 1$ cointegrating vector, the decomposition of Gonzalo and Granger (1995) identifies $k - r = 2$ common stochastic trends, $f_t = \alpha_\perp' Y_t$, where $\alpha_\perp$ spans the orthogonal complement of the adjustment vector. With the rational normalization that assigns unit weight to K_t^{NE} and r_t respectively, the estimated factors are

$$f_{1,t} = 0.12\, K_t^E + 1.00\, K_t^{NE}, \tag{S5.1}$$

$$f_{2,t} = -0.02\, K_t^E + 1.00\, r_t. \tag{S5.2}$$

The long-run representation of the system, $Y_t = A\, f_t + \mu_t$ with $A = \beta_\perp(\alpha_\perp' \beta_\perp)^{-1}$ and $\mu_t \sim I(0)$, has estimated loading matrix

$$\begin{pmatrix} K_t^E \\ K_t^{NE} \\ r_t \end{pmatrix} = \begin{pmatrix} 1.60 & 1.09 \\ 0.81 & -0.13 \\ 0.04 & 1.02 \end{pmatrix} \begin{pmatrix} f_{1,t} \\ f_{2,t} \end{pmatrix} + \begin{pmatrix} \mu_{1,t} \\ \mu_{2,t} \\ \mu_{3,t} \end{pmatrix}, \qquad \mu_t \sim I(0). \tag{S5.3}$$

The factor $f_{1,t}$ captures persistent coercive capacity (dominated by non-extortive kidnapping), while $f_{2,t}$ reflects permanent macro-financial conditions (dominated by the interest rate). The loadings of K_t^E (1.60 on f_1 and 1.09 on f_2) exceed unity, indicating that extortive kidnapping amplifies both permanent forces. The long-run trend of extortive kidnapping implied by the decomposition, $\tau_{K^E,t}^{\mathrm{VEC}} = 1.60\, f_{1,t} + 1.09\, f_{2,t}$, is the VEC–GG trend used in the equivalence exercise of Section S7.

S6. THE LOCAL LINEAR TREND MODEL

As an independent univariate contrast, extortive kidnapping is decomposed in state-space form (Harvey, 1989) into a stochastic level, a persistent slope, and a stationary cycle:

$$\begin{aligned} \text{Measurement:} \quad & y_t = \tau_t + c_t + \xi_t, && \xi_t \sim \mathcal{N}(0, \sigma_\xi^2), \\ \text{Trend:} \quad & \tau_t = \tau_{t-1} + b_{t-1} + \eta_t, && \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \\ \text{Slope:} \quad & b_t = \textstyle\sum_{j=1}^{8} \phi_j \, b_{t-j} + \zeta_t, && \zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2), \\ \text{Cycle:} \quad & c_t = \textstyle\sum_{k=1}^{5} \rho_k \, c_{t-k} + \kappa_t, && \kappa_t \sim \mathcal{N}(0, \sigma_\kappa^2), \end{aligned} \tag{S6.1}$$

estimated by maximum likelihood via the Kalman filter on the same monthly sample ($T = 858$, $\log L = -3{,}847.34$, $\text{AIC} = 7{,}728.68$, $\text{BIC} = 7{,}809.51$; 17 free parameters). Table V reports the full parameter set of the replication run. All four innovation standard deviations are significant at the 1% level ($p < 0.01$ on the log-scale parameters):

$$\hat{\sigma}_\eta \approx 7.13, \qquad \hat{\sigma}_\zeta \approx 3.51, \qquad \hat{\sigma}_\kappa \approx 6.77, \qquad \hat{\sigma}_\xi \approx 11.96. \tag{S6.2}$$

REMARK S6.1: Quasi-Newton maximum likelihood on this 15-state model can converge to slightly different points across MATLAB versions and starting values; the innovation standard deviations in (S6.2) are from the replication package run of `LLT_Extorsivo.m`. Small numerical differences do not affect any qualitative conclusion: in every run the permanent-shock variance σ_η^2 is large and highly significant.

The significance of $\hat{\sigma}_\eta$ at the 1% level confirms that permanent shocks dominate the long-run level of extortive kidnapping. Among the slope AR coefficients, ϕ_2, ϕ_3, ϕ_5, and ϕ_7 are significant at the 1% level; ϕ_8 is significant at the 10% level ($p = 0.06$) but not at the 5% level; ϕ_1, ϕ_4, and ϕ_6 are not significant at the 10% level. All five cycle AR coefficients (ρ_1 through ρ_5) are significant at the 1% level. The order-8 autoregressive structure of the slope captures gradual changes in the growth rate, consistent with slow adjustments in criminal capacity rather than abrupt breaks. The order-5 stationary cycle absorbs medium-run fluctuations without distorting the permanent component.

S7. EQUIVALENCE OF STOCHASTIC TRENDS

The VEC–GG trend of Section S5 and the LLT trend of Section S6 are built from conceptually independent principles: multivariate cointegration restrictions versus univariate state-space filtering. Their sustained coherence is the empirical manifestation of the following corollary, stated and proved as Corollary A1.2 in the proofs supplement (Appendix 2).

COROLLARY S7.1—Equivalence of stochastic trends: *Let $\{y_t\}$ be a time series integrated of order one, $I(1)$. Suppose the existence of two valid decompositions defined as*

$$y_t = P_t^{(1)} + u_t^{(1)} = P_t^{(2)} + u_t^{(2)},$$

where $P_t^{(1)}$ and $P_t^{(2)}$ are processes integrated of order one and $u_t^{(1)}, u_t^{(2)}$ are stationary $I(0)$ processes. Then

$$P_t^{(1)} - P_t^{(2)} \sim I(0).$$

TABLE V

LOCAL LINEAR TREND STATE-SPACE MAXIMUM-LIKELIHOOD ESTIMATES

Parameter	Estimate	Std. error	p-value	Sig.
$\log \sigma_\eta$ (level)	1.96	0.06	< 0.01	1%
$\log \sigma_\zeta$ (slope)	1.26	0.20	< 0.01	1%
ϕ_1 (slope AR)	−0.38	0.33	0.24	—
ϕ_2	−0.28	0.09	< 0.01	1%
ϕ_3	−0.51	0.16	< 0.01	1%
ϕ_4	−0.02	0.07	0.72	—
ϕ_5	0.50	0.05	< 0.01	1%
ϕ_6	−0.05	0.15	0.73	—
ϕ_7	0.67	0.11	< 0.01	1%
ϕ_8	0.59	0.31	0.06	10%
$\log \sigma_\kappa$ (cycle)	1.91	0.08	< 0.01	1%
ρ_1 (cycle AR)	−0.76	0.04	< 0.01	1%
ρ_2	−0.18	0.02	< 0.01	1%
ρ_3	−0.24	0.03	< 0.01	1%
ρ_4	−0.82	0.04	< 0.01	1%
ρ_5	−0.63	0.04	< 0.01	1%
$\log \sigma_\xi$ (irregular)	2.48	0.05	< 0.01	1%

Note. Kidnapping-for-ransom series; slope AR(8), cycle AR(5), stochastic level. Variance parameters estimated in logarithms. The Sig. column reports the highest conventional significance level (1%, 5%, 10%); "—" denotes not significant at 10%. *Source:* Authors' own calculations based on the monthly sample in Section S1 ($T = 858$). Software: MATLAB R2025b (The MathWorks, 2025); Econometrics Toolbox.

Consequently, all valid stochastic trends derived from the same observed series are equivalent up to a stationary component.

Empirically, the difference $d_t = \tau^{\mathrm{VEC}}_{KE,t} - \tau^{\mathrm{LLT}}_{KE,t}$ between the two estimated trends behaves as the corollary predicts. On the 1964–2025 display sample, the ADF test rejects a unit root in d_t at the 1% level ($p < 0.01$). The KPSS test with a Newey–West bandwidth of twelve lags does not reject level stationarity at the 5% level ($p = 0.06$) but would reject it at the 10% level ($p < 0.10$); with MATLAB's short default bandwidth the KPSS test rejects stationarity at the 5% level ($p = 0.01$), reflecting the persistence of d_t at short horizons rather than a stochastic trend. Following the corollary, the ADF result at the 1% level together with the KPSS(12) result at the 5% level support $d_t \sim I(0)$: the two estimates do not diverge systematically in the long run. The VEC through the Gonzalo–Granger decomposition and the LLT through agnostic state-space filtering independently reveal the same permanent component of the phenomenon, which makes the long-run characterization robust to the choice of econometric framework.

S8. RELEVANCE FOR THE MECHANISM

S8.0.0.1. *Scope.* This section explains how the macroeconometric results in Sections S1–S7 feed the mechanism-design model of the main paper (Section 2.2, "Macro dynamics: cointegration and the VEC foundation"). It is *interpretive*: it introduces no new estimates. Every empirical quantity cited below is reported in Sections S2–S7 and can be verified from the downloaded replication package using `README.txt` at the package root—without access to the author's working directories or any path outside the folder delivered to the reader.

S8.0.0.2. *Long-run equilibrium and the interest-rate channel.* The long-run equilibrium (S4.3) establishes that the interest rate is a structural determinant of extortive kidnapping: monetary contraction and credit restriction raise the cost of capital for criminal organizations, reshaping their incentives toward extraction strategies with faster returns ($\hat{\beta}_r/\hat{\beta}_{KE} \approx 1.31$ in the normalized relation). This macroeconomic environment is not neutral with respect to kidnapper types: it defines the support of the cost distribution and of the ransom valuation faced by heterogeneous agents in each period, feeding the parameter space within which the screening mechanism of the main paper operates.

S8.0.0.3. *Common trends and type heterogeneity.* The persistence of the two common trends identified in Section S5 (coercive capacity $f_{1,t}$ and macro-financial conditions $f_{2,t}$) implies that heterogeneity across types is neither transitory nor purely idiosyncratic, but structurally conditioned by the long-run macroeconomic environment; this property justifies its explicit treatment as a restriction on the type support in the mechanism-design model.

S8.0.0.4. *Trend equivalence and calibration anchors.* The trend-equivalence result of Section S7 closes the model-dependence argument: the permanent component is unique up to stationary perturbations (Corollary S7.1, proved in Appendix 2), so the long-run parameters carried into the calibration are structural anchors rather than artifacts of one econometric framework.

S8.0.0.5. *Reading guide for replicators.* Table VI maps each mechanism-relevant input to its location in this supplement and in the replication package. A reader who downloads the package can proceed as follows.

1. Read `tex/Appendix_1.pdf` (or this LaTeX source) through Section S7, then Section S8.
2. Open `README.txt` for folder layout and software requirements.
3. To *rerun* all estimates: execute Steps 1–3 in `README.txt` (MATLAB R2025b, Econometrics Toolbox).
4. To *inspect outputs without rerunning MATLAB*: open the pre-exported CSV files listed in the last column of Table VI; trend equivalence can be read from `matlab/Trend_Difference/trend_difference.csv` (column `Difference`).

Section S9 lists every script; the formal mechanism proofs and microeconometric priors (μ_0, hazards) are in the main paper and its other supplements, not in this package.

S9. REPRODUCIBILITY

All results in this supplement are reproduced by three MATLAB scripts included in the replication package that accompanies this document (folder `Appendix\_1/` when delivered with the thesis). No script references paths outside that folder. The VEC pipeline (unit roots, Johansen test with lag-sensitivity analysis, VEC estimation, Gonzalo–Granger decomposition, residual diagnostics, impulse responses, and forecasts) is implemented in `matlab/VEC/VEC_2T.m` with input data `matlab/VEC/Data.csv`; the estimated trends are exported to `matlab/VEC/Tendencias_SE.csv`. The LLT state-space model is implemented in `matlab/SE_Kalman/LLT_Extorsivo.m`, and the smoothed components (level, slope, cycle, residual) are exported to `matlab/SE_Kalman/LLT_decomposition.csv`. Trend equivalence (Section S7) is validated by `matlab/Trend_Difference/check_trend_equivalence.m`, which reads `matlab/VEC/Tendencias_SE.csv` and `matlab/SE_Kalman/LLT_decomposition.csv`. Section S8 (Table VI) explains how these outputs connect to the mechanism model. Scripts run on MATLAB R2025b with the Econometrics Toolbox; `README.txt` documents the replication steps for a reader working entirely from a local copy on their own computer.

TABLE VI

MAPPING OF MECHANISM INPUTS TO REPLICATED MACRO RESULTS

Mechanism input	Where in this supplement	Replication package
$I(1)$ order of K_t^E, K_t^{NE}, r_t	Section S2, Table I	`matlab/VEC/VEC\_2T.m` (Step 1 in `README.txt`)
Cointegration rank $r = 1$	Section S3, Table II	`matlab/VEC/VEC\_2T.m`
Long-run relation $K_t^E = 1.93\,K_t^{NE} + 1.31\,r_t + \varepsilon_t$	(S4.3), Section S4	`matlab/VEC/VEC\_2T.m`; β in console output
Adjustment: only K_t^E loads on the ECT (5% level)	Section S4, Table III	`matlab/VEC/VEC\_2T.m`
Common trends $f_{1,t}$, $f_{2,t}$ and VEC–GG loadings	Section S5, (S5.1)–(S5.3)	`matlab/VEC/VEC\_2T.m`; `matlab/VEC/Tendencias\_SE.csv`
LLT permanent component of K_t^E	Section S6, (S6.2), Table V	`matlab/SE\_Kalman/LLT\_Extorsivo.m`; `matlab/SE\_Kalman/LLT\_decomposition.csv`
Trend equivalence: $d_t = \tau_{K^E}^{\mathrm{VEC}} - \tau_{K^E}^{\mathrm{LLT}} \sim I(0)$	Section S7, Corollary S7.1	`matlab/Trend\_Difference/check\_trend\_equivalence.m`; `matlab/Trend\_Difference/trend\_difference.csv`
Full numeric check vs. all tables	Sections S2–S7	`matlab/Trend\_Difference/verify\_appendix1.m` (optional)

Note. This table is a reading guide only; it does not report new statistics. CSV files allow inspection of trends and differences without rerunning MATLAB. The mechanism model, calibration, and microeconometric priors are documented in the main paper. *Source:* Authors' own calculations based on the monthly sample in Section S1 ($T = 858$). Software: MATLAB R2025b (The MathWorks, 2025); Econometrics Toolbox.

REFERENCES

CNMH (2013), *Una sociedad secuestrada*, Imprenta Nacional, Bogotá. Centro de Memoria Histórica. [1]

CNMH (2026), "Portal de datos: El conflicto en cifras." Conflicto en Cifras. [1]

Dickey, David A. and Wayne A. Fuller (1979), "Distribution of the estimators for autoregressive time series with a unit root." *Journal of the American Statistical Association*, 74 (366), 427–431. [1]

Gonzalo, Jesus and Clive Granger (1995), "Estimation of common long-memory components in cointegrated systems." *Journal of Business & Economic Statistics*, 13 (1), 27–35. [4]

Harvey, Andrew C. (1989), *Forecasting, Structural Time Series Models and the Kalman Filter*, Cambridge University Press, Cambridge. [5]

Johansen, Soren (1988), "Statistical analysis of cointegration vectors." *Journal of Economic Dynamics and Control*, 12 (2–3), 231–254. Disponible en línea. [2]

Kwiatkowski, Denis, Peter C. B. Phillips, Peter Schmidt, and Yongcheol Shin (1992), "Testing the null hypothesis of stationarity against the alternative of a unit root." *Journal of Econometrics*, 54 (1–3), 159–178. [1]

Masih, Rumi (1995), "Modelling the dynamic interactions among crime, deterrence and socio-economic variables: evidence from a vector error-correction model." *Mathematics and Computers in Simulation*, 39 (3-4), 411–416. [1]

Ministerio de Defensa Nacional (2026), "Información estadística: Defensa y seguridad." Gobierno de Colombia, URL https://www.mindefensa.gov.co/irj/portal/Mindefensa. [1]

Suarez Padilla, Luis Carlos (2025), *Análisis Predictivo de Secuestro y Desaparición en Colombia: Un Enfoque con Series de Tiempo*. Tesis de especialización en estadística aplicada, Fundación Universitaria Los Libertadores, Bogotá, Colombia. Libertadores. [1]